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Enhancing daily peak water level forecasts in tidal-dominated regions using machine learning and future tide integration

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ABSTRACT

Tidal-dominated coastal urban areas, such as the Dong Nai-Sai Gon River Basin, face escalating flood risks from heavy rainfall, reservoir releases, and tides. Accurate daily peak water level (Z_{max}) forecasting is vital for flood and traffic management. Conventional harmonic analysis and standard machine learning (ML) methods often overlook future tidal data, limiting predictive accuracy. This study proposes a hybrid framework (M3) integrating 24 h forecasted tides, derived via Fourier analysis and a multilayer perceptron, with hydro-meteorological predictors using ensemble ML. Using data from four stations between 2019 and 2022, the optimal M3-cbr model achieved strong performance, yielding RMSE values from 0.0581 to 0.1103 m, KGE from 0.8499 to 0.9414, WI from 0.9509 to 0.9719, and |PBIAS| < 2.4325%. It improved accuracy by 9% to 43% over M2 and 45% to 86% over M1 during flood seasons. This cost-effective framework supports operational early warning and sustainable flood management.

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1 Introduction

Coastal urban areas are particularly vulnerable to natural disasters such as earthquakes, tsunamis, and flooding (Dai and Cai 2021). Urban flooding arises from the interplay of multiple extreme climatic events, including heavy precipitation, storms, extreme river discharge, and overload of natural systems (Zscheischler *et al.* 2020). Most research emphasizes rare extreme events that trigger flooding through storm surges, heavy rainfall, and high tides (Joyce *et al.* 2018, Bruneau *et al.* 2020, Dai and Cai 2021, Hasan Tanim and Goharian 2021, Chen *et al.* 2023, Shahabi and Tahvildari 2024). In low-lying coastal urban regions, frequent flooding, while not always catastrophic, can result in environmental pollution, transportation disruptions (Jacobs *et al.* 2018), and localized economic losses (Hino *et al.* 2019). For instance, Fant *et al.* (2021) report that high tides affecting transportation infrastructure in numerous United States coastal regions could escalate estimated annual damages from \$1.3–1.5 billion in 2020 to \$28–37 billion in 2050 and \$220–260 billion in 2100. Enhancing the precision of daily flood water level forecasts thus furnishes critical early warnings for flood control operations, proactive traffic management, and economic loss mitigation.

Tides represent a pivotal factor in flood forecasting for coastal urban zones. Conventional approaches predict water levels by decomposing harmonic constituents using Fourier

analysis, a technique initially proposed by William Thomson (1824–1907). Darwin (1893) advanced this technique using the equilibrium tide concept, which Doodson (1928) further refined by applying least-squares regression to derive harmonic constituents from extended observational water level records. Although capable of projecting tidal water levels at arbitrary future points, this method overlooks non-tidal components, yielding substantial errors near river mouths (Supharatid 2003).

An alternative paradigm involves hydraulic models (conceptual frameworks) that replicate physical processes by solving intricate differential equations. These models rely on gridded climatic inputs, projected tidal levels, and boundary flood discharges to simulate flows in sewer/river networks and overland runoff (Joyce *et al.* 2018, Hasan Tanim and Goharian 2021). Hydraulic models facilitate insights into physical mechanisms (Sivakumar *et al.* 2002), and can delineate flow velocity, inundation depth, and spatiotemporal flood extents. Nonetheless, they exhibit elevated uncertainty (Brandimarte and Di Baldassarre 2012, Oubennaceur *et al.* 2018), high computational demands, and long prediction durations, presenting substantial hurdles (Zhang *et al.* 2016, Zahura and Goodall 2022). Contemporary advancements seek to hybridize hydraulic models with machine learning, where the former computes water level discrepancies, and the latter is trained to anticipate these residuals (Tu *et al.* 2021, Fei *et al.* 2023, Dang *et al.* 2024).

While this synergy enhances precision, protracted computation and elevated costs render it ill-suited for rapid, cost-effective forecasting.

Contemporary investigations prioritize machine learning (ML) models to improve the accuracy of flood water level forecasts while ensuring efficient computation and minimal cost. Evidence suggests ML models frequently surpass hydraulic counterparts in representing non-linear and unstable dynamics. For instance, Zhang *et al.* (2016) introduced the SAPSO-BP framework, combining back-propagation (BP) neural networks with particle swarm optimization (PSO) to yield tidal water-level predictions superior to those of conventional harmonic techniques. Advancements in deep learning (DL) facilitate the modelling of intricate non-linear associations and superior extraction of long-range temporal dependencies, outperforming standard ML in predictive fidelity (Chen *et al.* 2021, Shahabi and Tahvildari 2024). For example, Shahabi and Tahvildari (2024) predict flood water levels by fusing convolutional neural networks (CNN) for spatiotemporal atmospheric data processing (wind speed, direction, and pressure) with long short-term memory (LSTM) for temporal dependency capture, training the model across spectra from quiescent marine conditions to major inundation episodes. Recent studies have further advanced ML applications in tidal-dominated regions (Rahman *et al.* 2025, Tran *et al.* 2025, Zhai *et al.* 2025, Zhou *et al.* 2025). While these works confirm the efficacy of hybrid data-driven methods in tidal environments, most still rely primarily on historical tidal constituents or decompose tidal signals rather than explicitly incorporating forecasted future tidal levels into the daily peak water level (Z_{max}).

ML/DL architecture demonstrates robust predictive performance for water levels, inundation depths, and storm surges, with fast execution and economical deployment. Prior efforts have predominantly leveraged ML/DL strengths to forecast floods in coastal metropolises during extreme events such as storms and deluges. Scant attention has been accorded to recurrent urban flooding, notwithstanding its substantial yearly fiscal toll. Only two studies address coastal urban inundation forecasting employing tidal, pluvial, and fluvial inputs (Bermúdez *et al.* 2019, Zahura *et al.* 2020). Zahura and Goodall (2022) recently developed an Random Forest (RF) model for estimating roadway submersion depth attributable to rainfall and tidal surges, augmenting ML inputs with road elevations and deriving outputs from Shen *et al.* (2019). Coastal urban flooding is recurrently governed by multifaceted influences encompassing stochastic (rainfall), deterministic (tides), and anthropogenic (reservoir discharge) factors, yet these remain understudied despite their salience. Formulating predictive models amid such synergies is imperative for anticipatory reservoir governance, alert dissemination for traffic orchestration, and consequent alleviation of quotidian inundation and cumulative regional detriment. Although numerous antecedents have developed frugal models, comprehensive reviews reveal no integration of prospective tidal water-level data. Conventional ML/DL paradigms predominantly leverage historical data for temporal water level prognoses. This reliance on historical records often fails to capture non-linear interactions with non-tidal disturbances, such as rainfall and reservoir

discharges. By incorporating forecasted tidal levels instead of average cycles, models can more accurately predict Z_{max} . Furthermore, tidal variances across river-to-sea transects in coastal metropolises modulate model efficacy but elude comprehensive appraisal.

Considering the limitations, particularly the predominant reliance on historical tidal data and the scant focus on non-extreme but recurrent urban flooding, the primary aim of this study is to develop and evaluate a novel hybrid ML framework. This framework uniquely integrates 24 h forecasted future tidal levels with non-tidal meteorological drivers to enhance the prediction of Z_{max} in complex, tide-dominated estuaries.

To achieve this aim, we formulate and systematically compare three progressive forecasting methodologies: method M1 (tidal baseline): utilizes tidal frequency components extracted via fast Fourier transform (FFT) in conjunction with multi-layer perceptron (MLP) to model the pure astronomical tide and forecast baseline hourly water levels. Method M2 (non-tidal forcing): applies ensemble ML models, including RF, Extreme Gradient Boosting (XGB), and Catboost regression (CBR). These models are trained exclusively on historical water levels, precipitation, and upstream reservoir discharges to capture non-tidal anomalies. The third approach, method M3 (hybrid integration), represents the proposed hybrid architecture, which explicitly feeds the forecasted tidal sequence derived from M1 into the ensemble models alongside the meteorological predictors. This integration allows the algorithms to learn the complex, non-linear interactions between deterministic tidal boundaries and stochastic hydro-meteorological noise.

By evaluating these models across four gauging stations distributed along a river-to-sea transect, this study provides a comprehensive assessment of spatial generalizability under varying tidal influences. Ultimately, the successful deployment of the M3 framework contributes to a cost-effective, high-precision tool for operational early warning systems, proactively assisting traffic coordination and reservoir governance in coastal metropolises experiencing daily inundation.

2 Materials and methods

2.1 Study area

The downstream region of the Dong Nai and Sai Gon (DNSG) river basin, encompassing key economic provinces of Vietnam such as Ho Chi Minh City, Long An, Dong Nai, and Binh Duong, was specifically selected as the study area due to its profound socio-economic significance and extreme hydrological complexity. The main river systems, including the Dong Nai, Be, Sai Gon, and Vam Co rivers, flow into the East Sea through the Soai Rap river mouth (Fig. 1). Hydrologically, this area is strongly influenced by the semi-diurnal tidal regime from the East Sea, with tidal amplitudes ranging from 3.0 to 4.0 m during spring tides and 1.8 to 2.0 m during neap tides.

Socio-economically, Ho Chi Minh City plays a central role in Vietnam's economic development, but the city's low-lying topography poses numerous challenges. According to statistics, 18.67% of the city's area has elevations below 0.5 m, 34.88% between 0.5–1.0 m, 22.13% between 1.0–2.0 m, and

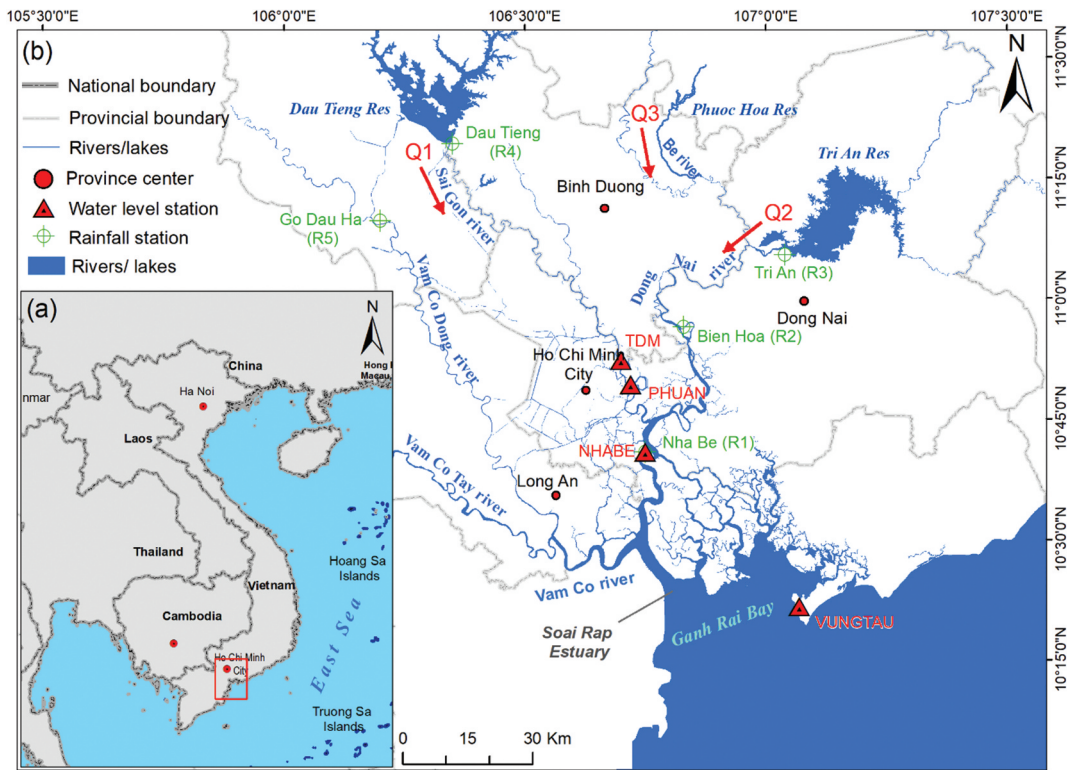


Figure 1. (a) Location of the study area within Vietnam, and (b) the Dong Nai–Sai Gon River Basin.

only 24.32% above 2.0 m. The low topography, combined with high population density and rapid urbanization, increases pressure on the drainage system, leading to frequent flooding when heavy rainfall coincides with high tides. Demonstrating the severity of this issue, national water level monitoring stations on the Sai Gon River, such as TDM and PHUAN, recorded extreme water levels of 1.75 m and 1.74 m, respectively, during the period from 2019 to 2022. These values significantly exceed the national alert level III threshold of 1.6 m. Ultimately, this intersection of powerful tidal forces, stochastic meteorological anomalies, and high socio-economic vulnerability makes this basin an ideal location to validate the proposed hybrid forecasting models and highlights the urgent need for effective early warning solutions.

2.2 Data sources and pre-processing

This study utilizes observational data from hydrological stations provided by the Ministry of Natural Resources and Environment (MONRE) from 2019 to 2022. Hourly water level data were collected at four monitoring stations located at varying distances from the Soai Rap river mouth, including TDM (86 km), PHUAN (74 km), NHABE (40 km), and VUNGTAU (coastal station), as detailed in Figs. 1 and 2(a) and Table 1. These hourly water level data are used directly for method M1, whereas Z_{max} derived from the hourly data is applied to methods M2 and M3. As described in Table 1, the average water level decreases progressively from river to sea, from 1.31 m at TDM to 0.84 m at VUNGTAU, while the std increases from 0.19 m at TDM to 0.24 m at VUNGTAU. This pattern reflects greater variability in near-sea areas, attributable to shallow water effects, waves, and wind. Although water

levels exhibit a relatively uniform distribution at most stations (with std < mean), the maximum values (ranging from 1.52 to 1.80 m) indicate severe flooding events in inner-city areas, particularly at NHABE and TDM.

Daily rainfall data were collected at five stations: NHABE (R1), BIENHOA (R2), TRIAN (R3), DAUTIENG (R4), and GODAUHA (R5), with locations shown in Fig. 1. The maximum daily rainfall ranges from 92.4 mm at R1 to 149.0 mm at R3, typically occurring during the rainy season (June to November), while the average rainfall varies from 4.39 mm at R1 to 7.03 mm at R4. The std is notably high, ranging from 10.96 to 15.54 mm, indicating substantial variability characterized by many days of no or light rain interspersed with heavy rainfall events (Fig. 2(b) and Table 1). As shown in Table 1, the interquartile range of rainfall appears extremely low (with the median equal to 0 mm and the 75th percentile at 2–6 mm). This skewed distribution is intrinsically linked to the distinct seasonal climate of the study area, which experiences an extended dry season with practically no precipitation. However, it is important to note that despite this low overall variability, highly concentrated rainfall events during the wet season are critical drivers of sudden increases in Z_{max} .

Discharge data from three reservoirs (Q1, Q2, and Q3) over the 2019–2022 period reveal significant differences in hydrological characteristics, reflecting distinct roles and operational strategies within the DNSG. The Dau Tieng Reservoir (Fig. 1) exhibits a low average discharge (20 m³/s) and limited variability (std of 42 m³/s), with a maximum of 204 m³/s, primarily serving water supply needs. In contrast, the Tri An Reservoir has the highest average discharge (433 m³/s), substantial variability (std of 308 m³/s), and a maximum discharge of 1960 m³/s, playing a critical role in power generation, flood regulation,

Table 1. Descriptive statistics of the input variables.

Statistical parameters	Period	Count	Mean	Standard deviation (std)	Minimum (Min)	0.25	0.5	0.75	Maximum (Max)
Daily rainfall (mm)									
R1	Whole	1461	4.39	10.96	0.00	0.00	0.00	2.20	92.40
	Training	1096	4.24	10.76	0.00	0.00	0.00	1.65	92.40
	Testing	365	4.87	11.52	0.00	0.00	0.00	3.40	85.40
R2	Whole	1461	5.41	13.10	0.00	0.00	0.00	4.00	138.60
	Training	1096	5.01	12.21	0.00	0.00	0.00	3.50	138.60
	Testing	365	6.62	15.40	0.00	0.00	0.10	6.30	135.90
R3	Whole	1461	5.36	13.13	0.00	0.00	0.00	4.00	149.00
	Training	1096	5.06	12.52	0.00	0.00	0.00	3.43	119.00
	Testing	365	6.28	14.78	0.00	0.00	0.00	5.80	149.00
R4	Whole	1461	7.03	15.54	0.00	0.00	0.00	6.00	139.60
	Training	1096	7.34	15.96	0.00	0.00	0.00	6.20	139.60
	Testing	365	6.11	14.18	0.00	0.00	0.00	5.00	125.00
R5	Whole	1461	4.77	12.84	0.00	0.00	0.00	2.20	121.00
	Training	1096	4.46	12.47	0.00	0.00	0.00	1.83	121.00
	Testing	365	5.71	13.86	0.00	0.00	0.00	4.00	112.00
Daily reservoir outflow (m ³ /s)									
Q1	Whole	1461	19.55	42.14	0.00	0.00	0.00	0.00	204.20
	Training	1096	4.20	18.37	0.00	0.00	0.00	0.00	200.00
	Testing	365	65.65	57.15	0.00	0.00	70.80	100.00	204.20
Q2	Whole	1461	433.43	308.48	0.00	150.00	368.00	683.00	1960.00
	Training	1096	405.84	313.20	0.00	140.00	320.00	631.00	1960.00
	Testing	365	516.25	278.23	0.00	242.00	564.00	785.00	983.00
Q3	Whole	1461	94.90	116.78	0.00	10.00	40.00	151.60	1069.70
	Training	1096	78.56	117.51	0.00	10.00	30.00	108.50	1069.70
	Testing	365	143.98	99.72	0.00	70.40	130.10	218.30	539.20
Daily peak water level (m)									
TDM	Whole	1461	1.31	0.19	0.65	1.17	1.31	1.45	1.75
	Training	1096	1.29	0.19	0.65	1.15	1.29	1.43	1.74
	Testing	365	1.39	0.18	0.93	1.25	1.39	1.53	1.75
PHUAN	Whole	1461	1.23	0.20	0.45	1.10	1.24	1.38	1.74
	Training	1096	1.22	0.20	0.45	1.08	1.21	1.36	1.74
	Testing	365	1.29	0.19	0.70	1.16	1.29	1.43	1.69
NHABE	Whole	1461	1.23	0.21	0.45	1.10	1.24	1.37	1.80
	Training	1096	1.21	0.21	0.45	1.08	1.22	1.34	1.80
	Testing	365	1.28	0.20	0.68	1.15	1.29	1.42	1.68
VUNGTAU	Whole	1461	0.84	0.24	−0.05	0.68	0.83	1.00	1.52
	Training	1096	0.83	0.22	0.15	0.68	0.82	0.98	1.52
	Testing	365	0.86	0.28	−0.05	0.68	0.85	1.07	1.43

and environmental flow maintenance. The Phuoc Hoa Reservoir shows intermediate values, with an average discharge of 95 m³/s, a std of 117 m³/s, and a maximum of 1070 m³/s, contributing to flood regulation and the sustenance of environmental flows (Fig. 2(b) and Table 1).

Prior to analysis, the data underwent pre-processing steps, including filling missing values, filtering, and normalization. The filtering method employed the interquartile range (IQR) technique to detect and handle outliers in the data columns. For each feature, the IQR was calculated as the difference between the third and first quartiles ($Q_{0.75} - Q_{0.25}$), with values outside the interval ($Q_{0.25} - 1.5 \times \text{IQR}$, $Q_{0.75} + 1.5 \times \text{IQR}$) identified as outliers. These outliers were replaced using linear interpolation to preserve data continuity. This approach ensures data cleanliness, minimizes the impact of anomalous values, and is well suited to developing ML models for water-level forecasting. Regarding normalization, we applied distinct strategies tailored to the inherent mathematical properties of each algorithm. For the MLP model, all input variables and target data were standardized using Z-scores, resulting in a mean of 0 and a std of 1. This approach was specifically chosen over strict boundary scaling to prevent mathematical saturation, thereby ensuring the network retains the capacity to predict unprecedented extreme water levels during the flood

season. Conversely, normalization was not applied to the RF, XGB, and CBR algorithms. Because these ensemble tree techniques partition data using specific threshold values, they are inherently scale-invariant and efficiently process raw, unscaled data. Following pre-processing, the data were divided into training and validation sets spanning 2019 to 2021, with the 2022 data reserved for testing.

2.3 Development of the ML-based forecasting models

Although numerous advanced techniques, including CNN-Transformer-SKANs, LSTM-based hybrids, and VMD-BPNN models, have recently been proposed for water level forecasting in tidal-dominated regions, the present study deliberately adopts a hybrid ensemble approach combining Fourier analysis with an MLP for tidal forecasting and tree-based ensemble ML (RF, XGB, and CBR) for non-tidal and hybrid modelling. This choice is driven by three practical considerations. First, the method requires only modest computational resources and performs robustly with limited historical data (four years), making it suitable for operational deployment in resource-constrained environments such as the DNSG basin. Second, ensemble tree-based models offer high interpretability and fast inference, which are essential for real-time daily peak water

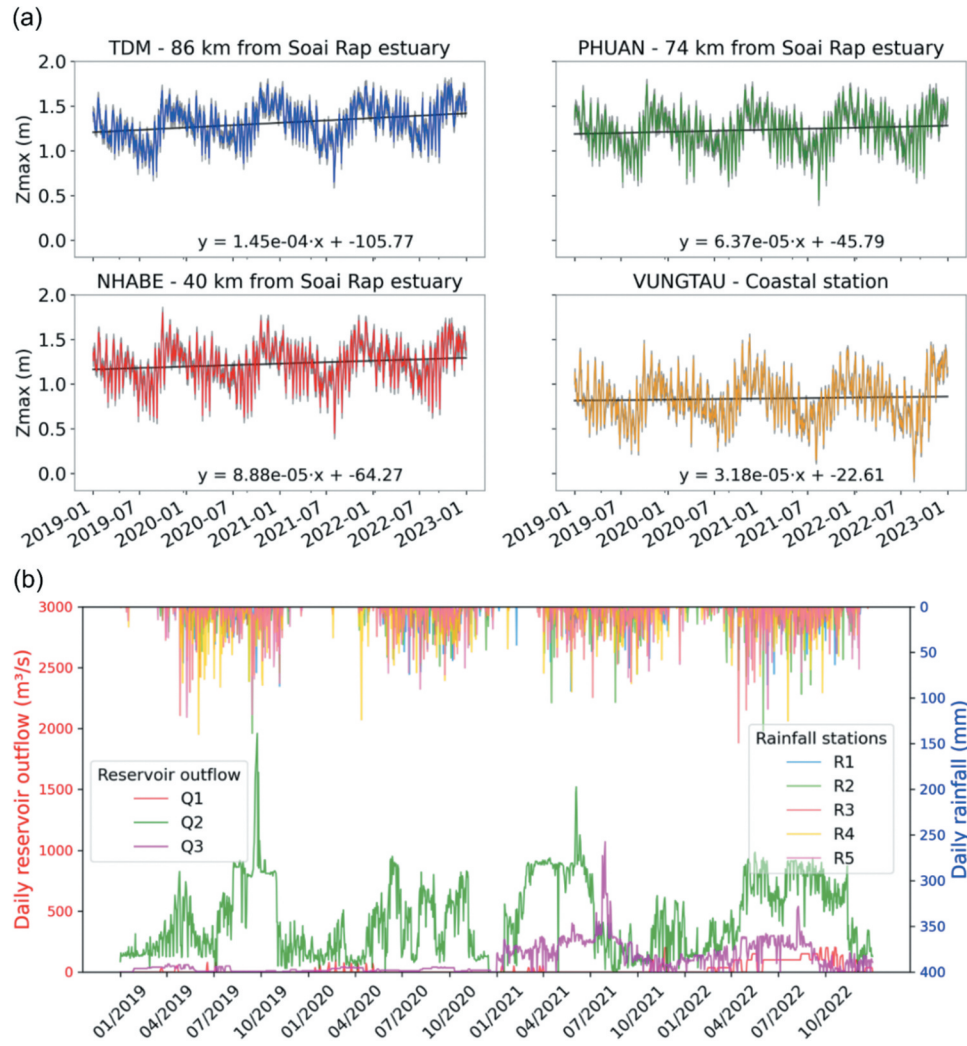


Figure 2. Hydrological data from 2019 to 2022: (a) observed peak water levels (Z_{max}) at monitoring stations; (b) corresponding upstream reservoir discharge and daily precipitation.

level warnings. Third, the framework uniquely enables seamless integration of 24 h forecasted future tidal levels into Z_{max} prediction, directly addressing the identified research gap in recurrent urban flooding rather than focusing solely on hourly or extreme-event forecasting.

In this study, we propose three novel methods to predict Z_{max} in tidal-influenced regions. The schematic in Fig. 3 illustrates how the three methods utilize distinct data sources for forecasting Z_{max} . Each method is trained with unique data strategies to highlight the superior performance improvements achieved by M3. Specifically, M1 uses only tidal frequency features, M2 uses non-tidal data (rainfall, reservoir discharges, and historical peak water levels), and M3 simultaneously uses both non-tidal data and the 24 h water levels at the forecasting time point calculated from M1. The data are pre-processed to remove noise and normalize values before being used to train the models. The detailed step-by-step algorithms and pseudocodes for models M1, M2 and M3 are provided in Algorithms (1) to (3) in the Supplementary material. The methods are described in detail as follows:

M1 combines Fourier analysis and MLP to forecast hourly water levels, from which Z_{max} is determined (Fig. 3(a)). This

method depends entirely on the fixed-frequency features of the dominant tidal constituents, which represent pure tidal information with cyclic characteristics that can be precisely determined at any future time point. The process involves the following steps: Step 1 employs Fourier analysis to examine hourly water levels at varying positions along the river-to-sea transect, thereby obtaining the frequencies of the dominant tidal constituents. Step 2 calculates the values $\cos(2\pi f_i t_j)$, $\sin(2\pi f_i t_j)$ at each forecasting time point t_j (s), where f_i is the frequency of the tidal constituent i . With n dominant constituents, this generates cyclic features at each t_j . Step 3 trains the MLP model with the features generated in Step 2 to predict hourly water levels. Step 4 uses the forecasted results over 24 h to determine Z_{max} each day.

M2 is entirely based on random non-tidal data, such as rainfall, discharges from reservoirs (Q1, Q2, Q3), and historical water level data, to forecast Z_{max} for the next day (Fig. 3(b)). This method avoids using tidal information, emphasizing non-cyclic factors such as rainfall and flows. It employs ensemble ML models based on decision trees, including RF (M2-rf), XGB (M2-xgb), and CBR (M2-cbr). These models are compared with M1 to assess the performance

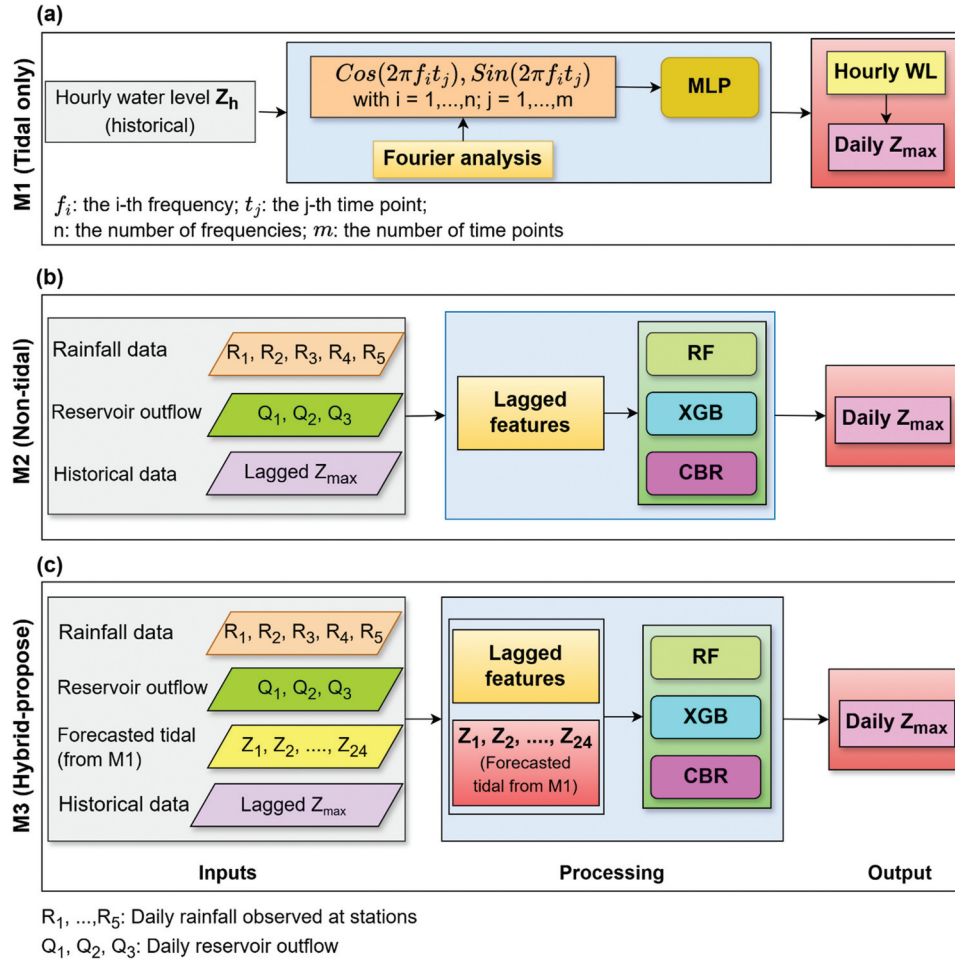


Figure 3. Schematic diagram illustrating the three forecasting methods (M1, M2, and M3) developed for $Z_{\max}$ prediction.

enhancements of M2. To identify the optimal ML model, this study incorporates lagged features corresponding to time intervals of 1, 2, and 3 days prior to the forecasting time point, capturing temporal influences on $Z_{\max}$. The generated features encompass rainfall at five stations (R_1, R_2, R_3, R_4, R_5), discharges from three reservoirs (Q_1, Q_2 and Q_3), and historical $Z_{\max}$ at the monitoring stations. Specifically, with a lag time of (k) days ($k = 1, 2, 3$), the features are generalized as follows: rainfall $R_i(t - k), R_i(t - k + 1), \dots, R_i(t)$ for each rainfall station $i = 1, 2, 3, 4, 5$; discharge $Q_j(t - k), Q_j(t - k + 1), \dots, Q_j(t)$ for each reservoir $j = 1, 2, 3$; historical peak water level: $Z_{\max}(t - k), Z_{\max}(t - k + 1), \dots, Z_{\max}(t - 1)$ at stations TDM, PHUAN, NHABE, VUNGTAU. For example, with (1-day lag), the features include $R_i(t - 1), R_i(t), Q_j(t - 1), Q_j(t)$, và $Z_{\max}(t - 1)$.

M3 is designed to forecast $Z_{\max}$ for the next day by integrating tidal data from M1 with non-tidal features, including rainfall at five stations (R_1, R_2, R_3, R_4 and R_5), discharges from three reservoirs (Q_1, Q_2 and Q_3), and historical $Z_{\max}$ (Fig. 3(c)). The tidal data are derived from M1's hourly water level forecasts over the next 24 h, based on Fourier analysis and the MLP model, and are represented by features $Z_t(h)$ (where $h = 1, 2, \dots, 24$). Ensemble ML models based on decision trees, RF (M3-rf), XGB (M3-xgb), and CBR (M3-cbr) are employed

due to their capacity to capture complex non-linear relationships between tidal and non-tidal features, particularly in highly variable conditions such as rainfall and discharges. Similar to method M2, M3 incorporates lagged features with time intervals of 1, 2, and 3 days, supplemented by the hourly forecasted water level features $Z_t(h)$ to enhance accuracy. M3 is compared with M1 and M2 to demonstrate its superior performance gains. For example, with $k = 1$, the features include $R_i(t - 1), R_i(t), Q_j(t - 1), Q_j(t), Z_{\max}(t - 1)$, và $Z_t(h)$ (với $h = 1, 2, 3 \dots 24$).

To comprehensively evaluate the complex hydrodynamics of the study area, the input variables were strategically structured into three progressive predictive scenarios (M1, M2, and M3), as detailed in Table 2. The basis for this selection is the isolation and integration of the distinct physical drivers of estuarine water levels. Specifically, method M1 was designed as a physics-guided baseline to capture the deterministic astronomical tide using Fourier analysis. Method M2 shifts the focus to non-tidal drivers, incorporating lagged rainfall and reservoir discharge to account for meteorological and upstream anthropogenic anomalies. Finally, method M3 represents the proposed hybrid framework. By explicitly feeding the 24 h forecasted tidal sequence from M1 (Z_1 to Z_{24}) into tree-based ML models alongside M2's non-tidal predictors, M3 bridges the gap between

Table 2. Input scenarios and their hydrological basis.

Input method	Features/variables included	Hydrological basis for selection
M1 (Tidal baseline)	Hourly harmonic constituents extracted via fast Fourier transform (FFT) of historical water levels: $\cos(2\pi f_i t)$, $\sin(2\pi f_i t)$	Astronomical tide: Isolates and models the deterministic, periodic cyclical nature of the tides. It provides the baseline water level of the estuarine system, excluding weather and upstream noise.
M2 (Non-tidal forcing)	Lagged daily predictors: Z_{max} , daily rainfall (R), daily reservoir outflow (Q)	Meteorological and upstream anomalies: Models the non-tidal disruptions. Lagged Z_{max} captures system memory, while rainfall and upstream reservoir discharges are the primary drivers of sudden, extreme pluvial and fluvial flood peaks during the flood season.
M3 (Hybrid integration)	- All M2 non-tidal predictors 24 h forecasted tidal sequence (Z_1 to Z_{24}) generated by M1	Integrated hydro-meteorological system: Provides the ML models (RF, XGB, CBR) with explicit knowledge of the future 24 h tidal curve. This allows algorithms to learn how non-tidal anomalies (rain/discharge) interact with specific tidal phases to accurately predict the ultimate Z_{max} .

deterministic tidal baseline constraints and non-linear pluvial/fluvial disruptions.

2.3.1 Fourier analysis

The Fourier analysis method was developed by Joseph Fourier in the early 19th century. It is considered a mathematical tool for analysing periodic signals into their basic wave components. In hydrology and environmental sciences, this method is commonly used to analyse time series such as tides or river water levels, helping to separate the tidal constituents. The Fourier series is represented as follows:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(2\pi n t / T) + b_n \sin(2\pi n t / T)) \quad (1)$$

where a_0, a_n, b_n are the Fourier coefficients calculated as in Equations (2)–(4):

$$a_0 = \frac{2}{T} \int_0^T f(t) dt \quad (2)$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos(2\pi n t / T) dt \quad (3)$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin(2\pi n t / T) dt \quad (4)$$

The Fourier transform converts the function $f(t)$ from the time domain to the frequency domain $F(\omega)$ as follows:

$$F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-i\omega t} dt \quad (5)$$

The inverse Fourier transforms to reconstruct the original signal from the frequency domain:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{i\omega t} d\omega \quad (6)$$

2.3.2 Machine learning models

To effectively capture the complex, non-linear interactions between tidal dynamics and hydro-meteorological factors, this study employed four ML algorithms: MLP, RF, XGB, and CBR. MLP serves as the foundational algorithm for method M1, specifically utilized to predict the harmonic astronomical tidal sequences. Meanwhile, RF, XGB, and CBR serve as the core predictive engines for Methods M2 and M3. To maintain a concise, focused narrative flow in the main

manuscript, the comprehensive theoretical backgrounds for all four algorithms are detailed in 'Appendix A. Readers interested in the fundamental mechanics of these models are directed to this appendix for further information.

2.3.3 Metrics for model performance evaluation

In this study, we evaluate the performance of the models in methods M1, M2, and M3 using the following evaluation metrics: root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2), percent bias (PBIAS), or Nash-Sutcliffe efficiency ($0 \leq NSE \leq 1$). The Kling-Gupta efficiency ($KGE \leq 1$), originally proposed by Gupta *et al.* (2009), and later refined by Kling *et al.* (2012), was developed to address certain limitations of NSE, particularly issues related to high bias and imbalanced variability. Therefore, KGE and Willmott's index of agreement ($0 \leq WI \leq 1$) were prioritized in identifying the optimal model. These metrics have been widely adopted in recent hybrid ML studies for groundwater level, river discharge, and evaporation forecasting (e.g. Samantaray *et al.* 2022, 2025, Samantaray and Sahoo 2023, Mahakur *et al.* 2025).

$$RMSE = \sqrt{(1/n) \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (7)$$

$$MAE = (1/n) \sum_{i=1}^n |(y_i - \hat{y}_i)| \quad (8)$$

$$NSE \text{ or } R^2 = 1 - \sum_{i=1}^n (y_i - \hat{y}_i)^2 / (y_i - \bar{y})^2 \quad (9)$$

$$PBIAS = \sum_{i=1}^n (y_i - \hat{y}_i) \times 100 / \sum_{i=1}^n y_i \quad (10)$$

$$KGE = 1 - \sqrt{(r - 1)^2 + (\alpha - 1)^2 + (\beta - 1)^2} \quad (11)$$

$$WI = 1 - \sum_{i=1}^n (\hat{y}_i - y_i)^2 / \sum_{i=1}^n (|\hat{y}_i - \bar{y}| + |y_i - \bar{y}|)^2 \quad (12)$$

where $y_i, \hat{y}_i, \bar{y}_i$ denote the observed value, predicted value, and mean of the observed value at time i , respectively; n represents the number of observations; r is the Pearson linear correlation coefficient between the observed and simulated values; the

variability ratio is defined as $\alpha = \sigma_s / \sigma_o$ (σ_s, σ_o represent the standard deviations of the simulated and observed time series); β is the bias ratio, defined as $\beta = \mu_s / \mu_o$ (denotes the mean values of the simulated and observed time series).

2.4 Hyperparameter optimization

In this study, the optimal parameters of the ML models were selected using the grid search method combined with three-fold cross-validation (grid search cross-validation, GCV-3) to minimize the risk of overfitting or underfitting. For method M1, the MLP model achieves optimal performance with the following parameters: activation function “tanh,” adaptive learning rate (learning_rate = adaptive), initial learning rate (learning_rate_init = 0.0005), and the hidden layer configuration detailed in Table B1 (Appendix B). Meanwhile, for methods M2 and M3, the RF, XGB, and CBR models are trained to leverage their stability against noise in the data, particularly suitable for rainfall and discharge features with high variability.

3 Results

3.1 Performance evaluation of M1

Method M1 uses the frequencies of dominant tidal constituents as important features in the hourly water-level forecasting model. However, the number of selected tidal constituents varies by location and significantly influences the model’s predictive performance. The evaluation results demonstrate that hourly water-level forecasting accuracy varies by location and is influenced by the number of dominant tidal

constituents selected (Table 3). At positions on the Sai Gon River (TDM and PHUAN), the best forecasting performance is achieved using 40 tidal constituents. The evaluation metrics RMSE, MAE, and R^2 for the validation set at the TDM station are 0.1733 m, 0.1371 m, and 0.9335, respectively. In contrast, the optimal number of tidal constituents for accurate predictions is 50 at the NHABE station on the Dong Nai River and 70 at the offshore VUNGTAU station. The statistical metrics (RMSE, MAE, and R^2) at NHABE are 0.2000 m, 0.1564 m, and 0.9532, respectively, while those at VUNGTAU are 0.1972 m, 0.1581 m, and 0.9496.

3.2 Performance comparison of methods M1, M2, and M3

The forecasting performance of all three methods for the 1-day lead time is summarized in Table 4, while the results for the 2- and 3-day lead times are presented in Tables S1 and S2 in the Supplementary material. At most stations, performance metrics (KGE, WI, NSE, PBIAS, RMSE) show minimal changes when adding 2–3-day lagged inputs, indicating limited benefit from longer rainfall and discharge lags. Therefore, the 1-day lag was adopted to reduce model complexity while maintaining accuracy.

As indicated in Table 4, forecasting quality varies both spatially and seasonally. Overall, the comparison across the four hydrological stations indicates clear performance differences among the three approaches. M3 (M3-rf, M3-xgb, and M3-cbr) consistently achieved the best performance, with KGE ranging from 0.8499 to 0.9414, WI from 0.9509 to 0.9719, NSE from 0.8306 to 0.890, $|PBIAS| < 2.4325\%$, and RMSE between 0.0581 and 0.1103 m. In comparison, M2 (M2-rf, M2-xgb, and M2-cbr) showed slightly lower

Table 3. Statistical errors (RMSE, MAE, and NSE) of hourly water levels using M1 (Note: Bold text indicates best performance).

		Number frequencies								
Metric	Period	8	16	20	30	40	50	60	70	80
Station: TDM										
RMSE (m)	Train	0.1843	0.1253	0.1217	0.1169	0.1097	0.1006	0.1026	0.0818	0.0801
	Validate	0.2302	0.1992	0.1892	0.1847	0.1733	0.1838	0.1797	0.1834	0.1810
MAE (m)	Train	0.1452	0.0985	0.0952	0.0911	0.0849	0.0780	0.0795	0.0627	0.0620
	Validate	0.1828	0.1593	0.1500	0.1454	0.1371	0.1477	0.1406	0.1463	0.1447
R ² (NSE)	Train	0.9243	0.9650	0.9670	0.9695	0.9731	0.9774	0.9765	0.9851	0.9857
	Validate	0.8828	0.9122	0.9208	0.9245	0.9335	0.9253	0.9285	0.9256	0.9275
Station: PHUAN										
RMSE (m)	Train	0.2586	0.1303	0.1266	0.1109	0.1014	0.1163	0.0821	0.1086	0.1052
	Validate	0.2874	0.2062	0.2013	0.1962	0.1942	0.1994	0.1960	0.1983	0.1980
MAE (m)	Train	0.2058	0.1010	0.0987	0.0857	0.0778	0.0897	0.0630	0.0841	0.0820
	Validate	0.2288	0.1640	0.1605	0.1542	0.1522	0.1546	0.1527	0.1548	0.1545
R ² (NSE)	Train	0.9085	0.9768	0.9781	0.9832	0.9859	0.9815	0.9908	0.9839	0.9849
	Validate	0.8918	0.9443	0.9469	0.9496	0.9506	0.9479	0.9497	0.9485	0.9487
Station: NHABE										
RMSE (m)	Train	0.2617	0.1491	0.1385	0.1294	0.1048	0.1199	0.1163	0.1114	0.1103
	Validate	0.2974	0.2199	0.2084	0.2215	0.2203	0.2000	0.221	0.2037	0.2619
MAE (m)	Train	0.2081	0.1154	0.1068	0.0988	0.0794	0.0926	0.0889	0.0845	0.0842
	Validate	0.2375	0.1747	0.1656	0.175	0.1741	0.1564	0.1765	0.1598	0.2095
R ² (NSE)	Train	0.9132	0.9718	0.9757	0.9788	0.9861	0.9818	0.9829	0.9843	0.9846
	Validate	0.8966	0.9434	0.9492	0.9426	0.9433	0.9532	0.9429	0.9515	0.9198
Station: VUNGTAU										
RMSE	Train	0.2576	0.1549	0.1552	0.0905	0.0790	0.0744	0.0719	0.0693	0.0687
	Validate	0.3027	0.2026	0.2048	0.2287	0.2177	0.2073	0.2048	0.1972	0.2227
MAE	Train	0.1995	0.1037	0.1052	0.0682	0.0591	0.0556	0.0537	0.0517	0.0515
	Validate	0.2461	0.1609	0.1634	0.1763	0.1727	0.1648	0.1646	0.1581	0.1777
R ² (NSE)	Train	0.9058	0.966	0.9658	0.9884	0.9911	0.9922	0.9927	0.9932	0.9933
	Validate	0.8811	0.9467	0.9456	0.9321	0.9385	0.9442	0.9455	0.9496	0.9356

Table 4. Metrics (KGE, WI, NSE, PBIAS, and RMSE) at four stations for the 1-day lead time.

Model	1-day lead time									
	Dry season					Flood season				
	KGE	WI	NSE	PBIAS	RMSE	KGE	WI	NSE	PBIAS	RMSE
Station: TDM										
M1	0.7053	0.8066	0.2929	4.2007	0.1240	0.8513	0.8890	0.5595	6.1300	0.1321
M2-rf	0.8917	0.9472	0.8066	0.1652	0.0648	0.9015	0.9566	0.8428	0.6431	0.0789
M2-xgb	0.8910	0.9469	0.8052	0.0326	0.0651	0.9076	0.9544	0.8329	1.2789	0.0813
M2-cbr	0.8868	0.9475	0.8105	0.2603	0.0642	0.8886	0.9557	0.8433	1.0593	0.0788
M3-rf	0.8906	0.9548	0.8395	0.7621	0.0591	0.9315	0.9686	0.8835	1.3819	0.0679
M3-xgb	0.8829	0.9577	0.8522	1.1439	0.0567	0.9371	0.9672	0.8775	2.1383	0.0697
M3-cbr	0.8499	0.9520	0.8387	1.1241	0.0592	0.9358	0.9690	0.8846	1.9221	0.0676
Station: PHUAN										
M1	0.7173	0.8547	0.3740	3.9520	0.1237	0.8238	0.9156	0.6483	5.4251	0.1310
M2-rf	0.8736	0.9369	0.7735	0.6709	0.0744	0.8826	0.9462	0.8078	0.7782	0.0968
M2-xgb	0.8645	0.9322	0.7612	1.3437	0.0764	0.8841	0.9457	0.8062	1.3071	0.0972
M2-cbr	0.8752	0.9370	0.7737	0.8411	0.0744	0.8816	0.9491	0.8202	0.9791	0.0937
M3-rf	0.8564	0.9531	0.8403	0.9142	0.0625	0.9246	0.9659	0.8740	1.2357	0.0784
M3-xgb	0.8673	0.9585	0.8571	0.9653	0.0591	0.9398	0.9716	0.8933	1.3110	0.0721
M3-cbr	0.8676	0.9598	0.8620	0.9569	0.0581	0.9383	0.9719	0.8945	1.1185	0.0717
Station: NHABE										
M1	0.7193	0.8580	0.4168	1.3210	0.1168	0.8519	0.9303	0.7112	5.4464	0.1211
M2-rf	0.8649	0.9330	0.7595	0.3781	0.0750	0.8840	0.9410	0.7848	0.7915	0.1046
M2-xgb	0.8521	0.9295	0.7490	−0.7992	0.0766	0.8689	0.9412	0.7922	0.1931	0.1027
M2-cbr	0.8287	0.9308	0.7656	−0.5521	0.0740	0.8263	0.9407	0.8038	−0.0471	0.0999
M3-rf	0.8684	0.9460	0.8112	0.6591	0.0664	0.9317	0.9641	0.8642	1.5423	0.0831
M3-xgb	0.8919	0.9517	0.8264	0.9295	0.0637	0.9449	0.9705	0.8871	1.8586	0.0758
M3-cbr	0.8653	0.9509	0.8306	0.6233	0.0629	0.9414	0.9714	0.8924	1.7640	0.0739
Station: VUNGTAU										
M1	0.8141	0.9080	0.6331	1.0348	0.1145	0.8468	0.9358	0.7887	4.9043	0.1595
M2-rf	0.8860	0.9491	0.8167	−0.0064	0.0810	0.8493	0.9639	0.8763	−1.1992	0.1220
M2-xgb	0.8755	0.9453	0.8041	−0.6043	0.0837	0.8414	0.9624	0.8725	−1.1133	0.1239
M2-cbr	0.8486	0.9488	0.8250	−0.7107	0.0791	0.7688	0.9563	0.8621	−2.4746	0.1289
M3-rf	0.8781	0.9660	0.8812	0.3367	0.0652	0.8537	0.9689	0.8956	1.1412	0.1122
M3-xgb	0.9009	0.9625	0.8655	0.9752	0.0694	0.8762	0.9677	0.8903	3.4578	0.1149
M3-cbr	0.8995	0.9653	0.8760	1.0481	0.0666	0.8609	0.9699	0.8990	2.4325	0.1103

performance, with RMSE values more than 8% higher despite similar KGE, WI, NSE, and PBIAS values, whereas M1 performed the worst.

When results are separated by season (dry and wet), the performance of the approaches varies considerably. During the dry season, M3 and M2 remain the best-performing approaches, with RMSE values ranging from 0.0567 to 0.0837 m, whereas M1 exhibits higher errors (RMSE = 0.1145–0.1240 m). In the wet season, the RMSE of all approaches increases; however, M3 still demonstrates superior performance, with RMSE reductions of 9–43% compared to M2 and 45–86% compared to M1.

From a spatial perspective, moving from upstream towards the coastal area, the performance of the approaches gradually decreases. The VUNGTAU station shows the lowest performance among the four stations (TDM, PHUAN, NHABE, and VUNGTAU). Nevertheless, M3 still maintains strong performance, with KGE > 0.8499, NSE > 0.8112, and RMSE < 0.1149 m.

For enhanced visualization, a radar chart (Fig. 4) compares multimetric (KGE, WI, and NSE) performance, while the Taylor diagram (Fig. 5) provide insights into error distribution and model-observation agreement, respectively. Notably, the M3-cbr and M2-cbr models deliver the highest performance in both dry and flood seasons. Accordingly, M1, M2-cbr, and M3-cbr were selected as representative models for comparative analysis.

Figure 6 depicts the distribution of Z_{max} from the three models (M1, M2-cbr, M3-cbr) and observed values. The distributions differ markedly between the dry and flood seasons. During the dry season, Z_{max} most frequently occurs in the 1.2–1.3 m range at river stations (TDM, PHUAN, and NHABE) and 0.7–0.8 m at the offshore VUNGTAU station. Model M3-cbr shows a data distribution that closely aligns with the observed values, surpassing M1 and M2-cbr. In the flood season, values in the 1.5–1.7 m range, corresponding to alert levels II–III (1.5–1.6 m) and causing widespread flooding, are most common at TDM, PHUAN, and NHABE stations, while 1.2–1.4 m predominates at VUNGTAU. Model M3-cbr again exhibits superior performance relative to M1 and M2-cbr. Notably, for water levels exceeding 1.8 m (indicating severe flooding beyond alert level III at 1.6 m), M3-cbr forecasts align closely with observations. Model M1 performs the worst, with a median value substantially below the observations and an inability to realistically replicate extreme values. Although M2-cbr performs better, it falls short of M3-cbr in reproducing extremes. Figure 7 illustrates the variation in predictive accuracy across the three representative methodologies (M1, M2, and M3). The empirical results clearly indicate a progressive improvement in model performance. M1 yields the lowest overall accuracy for predicting extreme peaks. M2 shows a notable improvement over method M1 by incorporating additional variables. Ultimately, M3

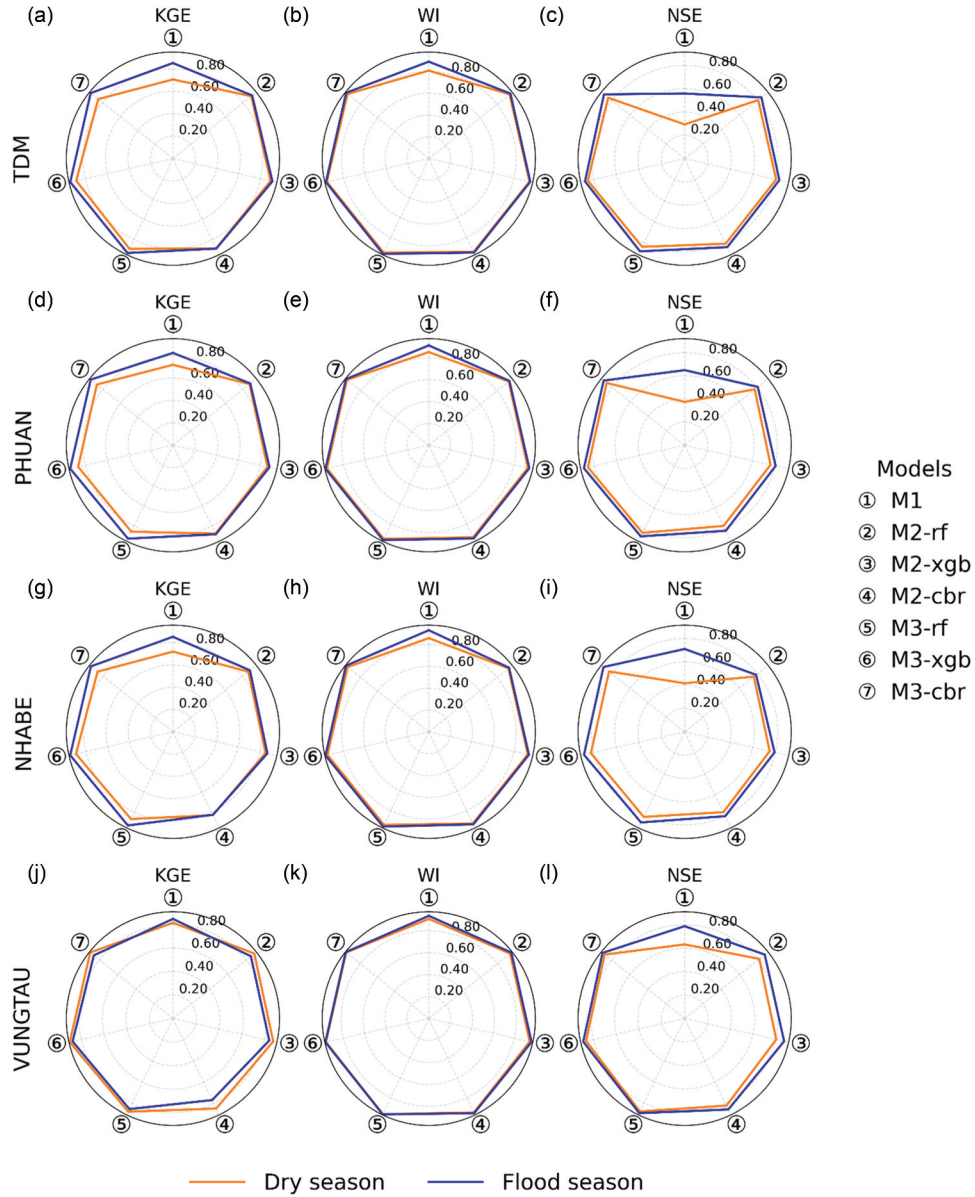


Figure 4. Radar chart comparing peak water level forecasting performance (KGE, WI, and NSE) across four stations.

achieves the highest accuracy and the lowest error metrics among all evaluated frameworks.

4 Discussion

Our study demonstrates that integrating 24 h tidal water level data, forecasted 1 day in advance, substantially enhances the M3 model's water level forecasting performance, surpassing methods M1 and M2. This finding corroborates the study's initial hypothesis: incorporating deterministic information at future time points, such as tidal water levels, can improve the accuracy of peak water-level predictions in tide-influenced regions. These results not only validate the hypothesis but also pave the way for innovative applications of tidal data in developing cost-effective flood forecasting models for urban areas subject to tidal effects.

To provide a deeper understanding of the model's decision-making process, we conducted SHapley Additive exPlanations

(SHAP) analysis on the best-performing M3-cbr model at the representative NHABE station (Fig. 8). The results show that the previous day's Z_{max-1} is the most influential feature, followed closely by the 24 h forecast tidal levels (Z_1 to Z_{24}), while non-tidal factors (rainfall and reservoir discharges) exert a comparatively smaller but still meaningful influence. Critically, this feature ranking is highly consistent with the physical processes governing tide-dominated river systems: strong short-term persistence (autocorrelation) combined with deterministic tidal forcing largely controls daily peak water levels, whereas rainfall and reservoir operations act mainly as stochastic modulators. The prominent role of forecasted tidal levels directly supports the central hypothesis of this study that incorporating future tidal information substantially improves Z_{max} prediction beyond what historical data alone can achieve. However, the relatively modest contribution of rainfall suggests a limitation of using daily-aggregated predictors in a tropical urban setting, where intense, short-

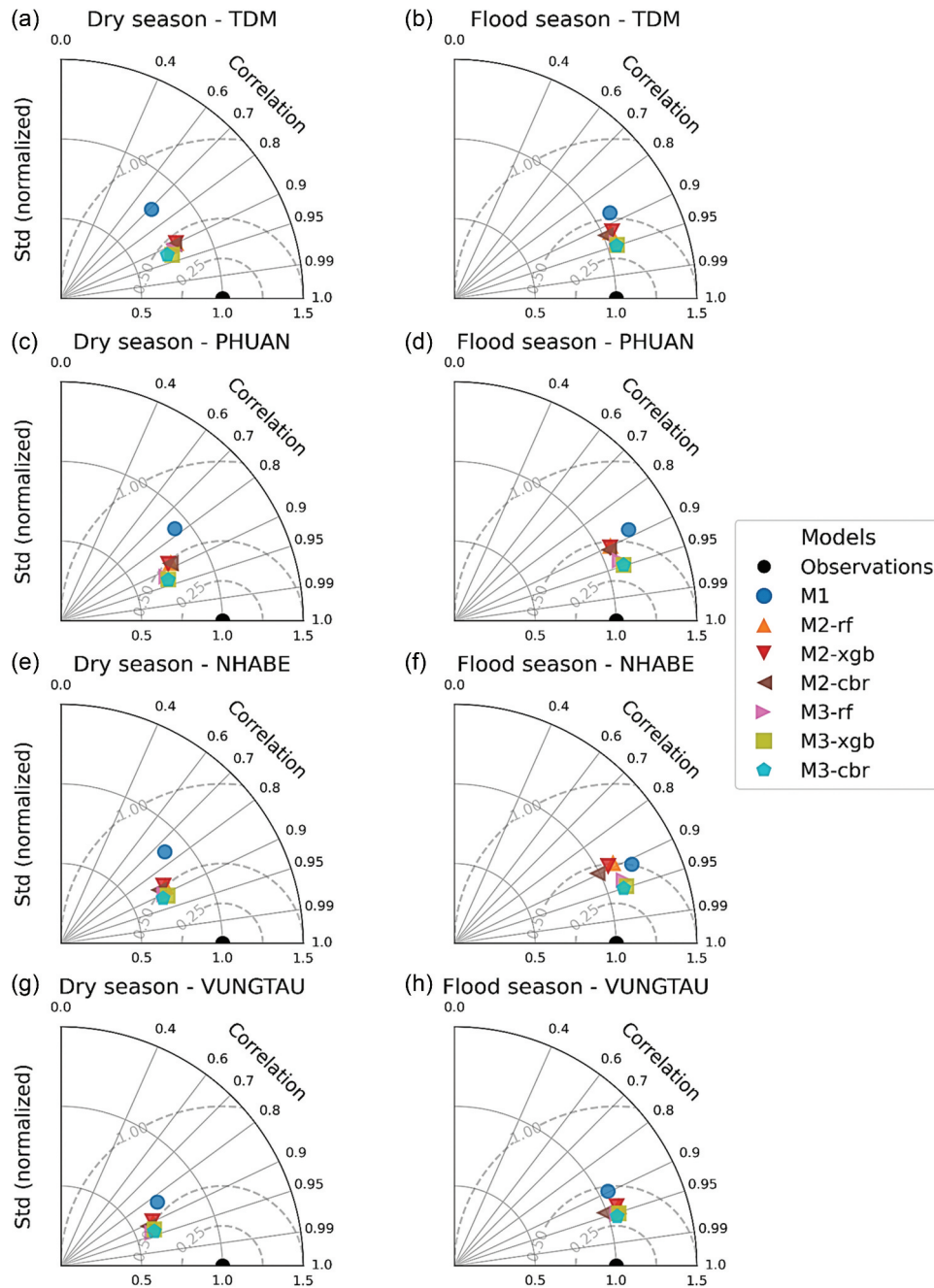


Figure 5. Taylor diagram illustrating model performance across correlation, centred RMSE, and standard deviation relative to observations at four stations.

duration rainfall events can trigger rapid flooding; finer temporal resolution or additional sub-daily rainfall features may further enhance model performance under extreme conditions. From a methodological viewpoint, although SHAP offers excellent global and local interpretability for tree-based ensembles, it relies on the assumption of feature independence and may underestimate complex non-linear interactions (e.g. the synergistic effect between high tides and heavy rainfall during spring tides). This highlights the value of combining SHAP with complementary explainable AI techniques, such as partial dependence plots or SHAP interaction values, in future work. Overall, the analysis not only confirms the effectiveness and physical plausibility of the proposed hybrid M3 framework but also provides practical guidance for operational flood

forecasting: prioritizing high-accuracy tidal forecasts and continuous historical water-level monitoring will deliver the greatest improvements in predictive skill for recurrent urban inundation in tidal-dominated coastal cities.

Compared with state-of-the-art studies, the proposed M3 framework offers distinct advantages for operational Z_{max} forecasting in tidal-dominated urban rivers. While advanced DL models such as CNN-Transformer-SKANs have achieved excellent hourly accuracy ($RMSE < 0.03$ m, $NSE \approx 0.99$) in coastal lagoon settings (Rahman *et al.* 2025), these approaches typically require high-resolution meteorological inputs and substantial computational resources, limiting their applicability in data-scarce developing regions. Similarly, recent hybrid models in estuarine regions have successfully reduced

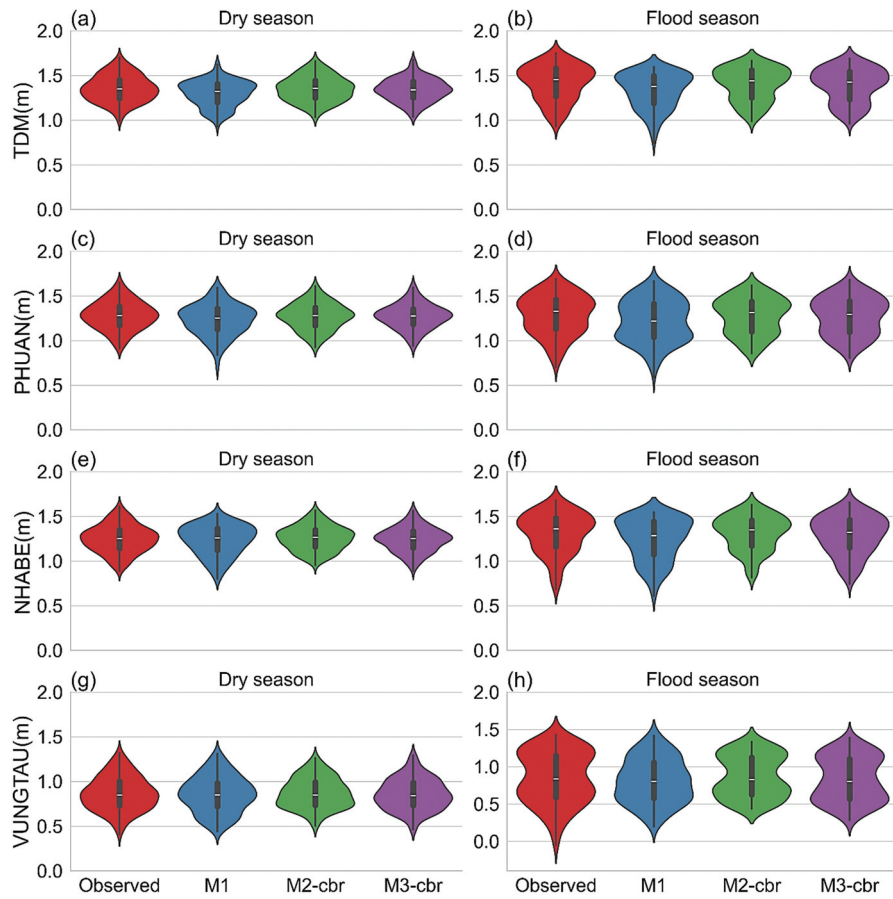


Figure 6. Violin plots comparing the performance of three methods at four stations.

prediction errors (Tran *et al.* 2025). However, many of these approaches still rely primarily on harmonic constituents derived directly from historical data, rather than the explicit integration of projected future tidal forecasts. This highlights a critical gap that our proposed framework aims to address. Previous studies on flood forecasting, such as Shen *et al.* (2019), Zahura *et al.* (2020), and Zahura and Goodall (2022), combine physical models and ML to predict detailed parameters, including flow velocity and flood depth in urban roadways. However, these approaches entail high costs, extended computation times, and complex data requirements, limiting their widespread application. By contrast, our M3-cbr model delivers high accuracy using modest computational resources and daily data. It achieved RMSE values of 0.0581–0.0666 m during the dry season and 0.0676–0.1103 m during the flood season. These results specifically target the critical daily maximum water level, which is far more relevant for flood warnings and traffic management than hourly averages. This confirms that the explicit integration of 24 h forecasted tidal levels addresses a key gap in the literature, where most recent ML studies (Tran *et al.* 2025, Zhai *et al.* 2025, Zhou *et al.* 2025) either decompose tidal signals or use only historical data, resulting in reduced skill during peak inundation events driven by non-linear rainfall–tide–discharge interactions. Furthermore, M3 exhibits significant improvement compared to a similar study by Islam *et al.* (2025) in the coastal region of Bangladesh, where the RMSE for Z_{max} forecasting ranged from

0.64 to 0.77 m. In contrast, our study develops a low-cost ML model for forecasting daily peak river water levels, enabling flood warnings in urban centres without relying on intricate physical models. By comparing Z_{max} against alert thresholds, this method streamlines the forecasting process, reducing both costs and computational demands. Moreover, seasonal interactions between river discharge and tidal dynamics induce tidal distortion, affecting water levels throughout the river from upstream to the estuary and thereby altering forecasting performance across locations. Through model evaluation across multiple sites from river to sea during both dry and flood seasons, we demonstrate that ML methods can leverage tidal data to enhance explanatory power, achieving accuracy comparable to or superior to traditional methods while offering greater simplicity and cost-effectiveness.

4.1 Influence of seasonal and tidal variations on M1, M2, and M3 performance

Consequently, the low annual statistical variability of rainfall inherently reduces its aggregate importance score. Nevertheless, standard astronomical tide calculation methods are purely deterministic and cannot account for meteorological anomalies. Therefore, the methodological approach of our framework rests not on reiterating the dominance of tides, but rather on its ability to successfully isolate and model non-tidal noise, specifically intense seasonal rainfall. By integrating ML

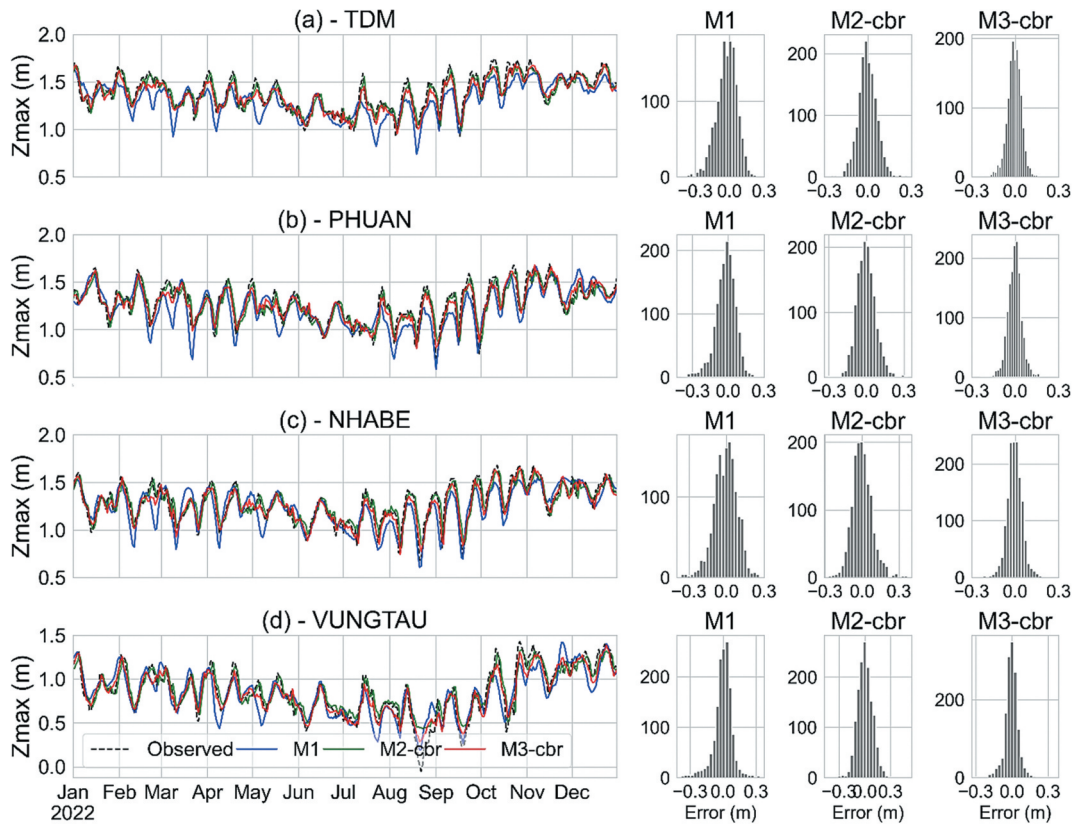


Figure 7. Daily peak levels and error distributions of three methods at four stations.

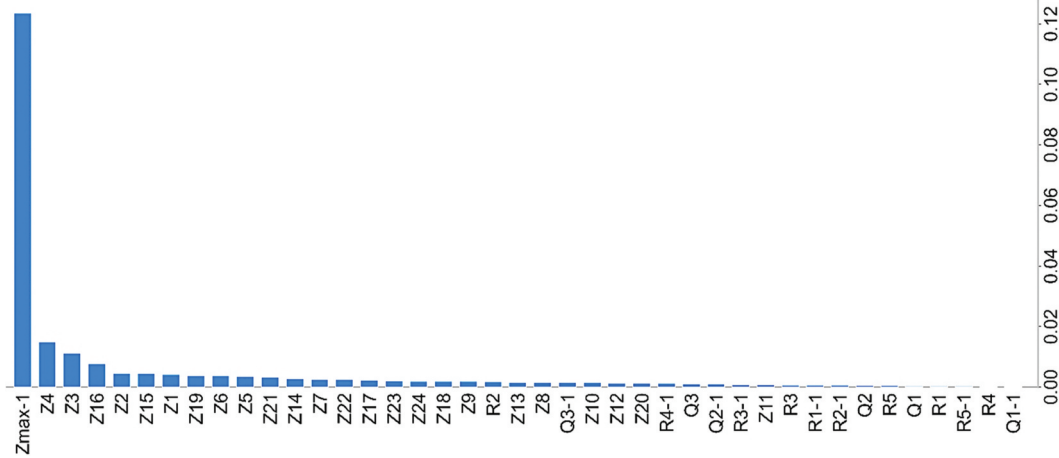


Figure 8. SHAP feature importance analysis for M3-cbr at the representative station NHABE.

to capture these rainfall-induced anomalies, the model significantly enhances the prediction of Z_{max} , which traditional tide-only methods frequently underestimate.

Model forecasting performance varies spatially from inland river reaches to marine environments. RMSE tends to increase progressively from inland rivers to the sea, whereas model generalization KGE, WI, and NSE improve in the opposite direction. This escalating RMSE reflects heightened data variability and complexity driven by multifaceted factors, including bottom topography, vegetation cover (e.g. mangrove forests), and wave-induced tidal distortion, which collectively complicate predictions. Conversely, generalization is stronger

offshore, as models can extract key relationships from dynamic data, such as tidal cycles, facilitating accurate forecasts on novel datasets. In inland rivers, data exhibit lower variability but may capture only part of the complexities, thereby diminishing generalization. This pattern aligns with prior research, which observes higher RMSE in coastal zones due to inherent complexity, yet notes enhanced generalization through learned relational dynamics (Li *et al.* 2020b, Vizi *et al.* 2023).

Overall, RMSE is lower during the dry season than in the flood season, though generalization KGE, WI, NSE is comparatively weaker in the former. The downstream DNSG basin is predominantly influenced by East Sea tides (Le *et al.* 2022,

2025b), making tidal effects more dominant during floods than non-tidal inputs such as rainfall and upstream reservoir discharges. This dominance helps models capture water-level trends, despite elevated absolute errors due to amplified amplitudes. By contrast, dry-season water levels are more stable and less prone to extremes (e.g. heavy rainfall), yielding reduced variability and thus lower RMSE. However, such stability may induce overfitting, limiting model efficacy to analogous conditions and constraining applicability in scenarios like flood seasons. Although the primary challenges arise during the flood season, the M3 approach, particularly the optimal M3-cbr model, consistently achieves the lowest RMSE compared to the other models. The substantial variation in predictive accuracy observed in Fig. 7 is primarily governed by how each method assimilates these physical boundary conditions and handles the severe hydro-meteorological noise during the flood season. M3 significantly outperforms M1 and M2 by directly resolving their individual blind spots regarding isolated data inputs. It achieves this superiority by simultaneously feeding the forecasted tidal sequence for the upcoming 24 h and meteorological predictors into the algorithm. Using the tide as a deterministic baseline, method M3 successfully isolates stochastic noise from rainfall and reservoir outflows to retrieve perfectly aligned historical analogues, thereby drastically reducing prediction errors.

4.2 Limitations and future outlooks

The proposed M3 hybrid framework offers several key advantages for Z_{max} forecasting in tidal-dominated urban regions. First, it achieves superior accuracy (e.g. RMSE decreased by 9–43% compared with M2, and by 45–86% compared with M1) by uniquely integrating 24 h forecasted tidal levels with non-tidal drivers, outperforming traditional methods that rely solely on historical data. Second, the approach is cost-effective and computationally efficient, requiring only modest resources and readily available data (e.g. 4 years of observations), making it suitable for operational deployment in resource-constrained developing countries like Vietnam. Third, its ensemble ML backbone (CBR) provides high interpretability via SHAP analysis, enabling stakeholders to understand key drivers such as lagged water levels and tidal fluctuations for informed flood management.

Although this study yields promising results in forecasting peak water levels using low-cost ML models, several limitations warrant attention. First, the dataset spanning 2019 to 2022 lacks sufficient duration to capture extreme flood discharge events, resulting in limited water-level fluctuations at sites along the Sai Gon River and potentially compromising the model's predictive accuracy under severe conditions. Second, variations in peak water levels are influenced by alterations in river basin bed characteristics, such as urbanization or shifts in vegetation cover, yet these factors were not incorporated, hindering the model's ability to elucidate relationships between urban development and peak water dynamics. Finally, the investigation centres on the DNSG region, characterized by the interplay of tides, rainfall, and reservoir discharges; its generalizability to other locales with distinct hydrological profiles remains unverified.

In addition to these limitations, uncertainties inherent in the modelling process must be acknowledged. Data uncertainties arise from potential inaccuracies in observational inputs, such as rainfall measurements or reservoir discharge records, which could propagate errors in model outputs (Brandimarte and Di Baldassarre 2012). Model uncertainties, including those related to parameter selection and overfitting in ML algorithms, may further amplify prediction variance, particularly in hybrid approaches that blend physical and ML components (Bentivoglio *et al.* 2022). Addressing these uncertainties through sensitivity analyses or ensemble modelling could strengthen future iterations.

To mitigate these limitations and uncertainties while amplifying the study's impact, several avenues for future research merit exploration. First, validating the model across diverse riverine and coastal settings would assess its generalizability and enable adaptations to varied hydrological contexts. Second, incorporating data on evolving river basin bed attributes, such as urbanization extent or vegetation modifications, would bolster the model's explanatory capacity for drivers of peak water levels, thereby informing sustainable land-use planning. Third, deploying the model within real-time flood warning systems integrated with monitoring networks would enhance its practical utility, particularly for flood mitigation. Finally, embedding climate change scenarios would safeguard the model's efficacy amid evolving environmental conditions. Additionally, integrating advanced DL architectures (e.g. LSTM hybrids) with high-resolution real-time data (e.g. IoT sensors or satellite imagery) could address sub-daily extremes and long-term dependencies, further enhancing predictive skill under climate variability.

5 Conclusions

This study presents a novel hybrid ML framework designed to enhance the forecasting accuracy of Z_{max} in complex, tidally dominated estuaries, specifically in the downstream basin of the Dong Nai and Sai Gon river systems. The core methodological novelty lies in the proposed method M3, which explicitly bridges deterministic boundaries with empirical algorithms. It achieves this by integrating forecasted astronomical tidal sequences for the upcoming 24 h (derived via Fourier analysis and a multilayer perceptron in method M1) with meteorological and anthropogenic drivers independent of the tide (method M2) into advanced ensemble models.

Rigorous evaluation along a spatial transect from the river to the sea yielded robust conclusions about the predictive dynamics of the framework. The results unequivocally demonstrate that the hybrid M3 approach substantially outperforms standalone tidal or meteorological models. Notably, at stations nearest the river mouth with intense tidal dominance (such as Vung Tau and Nha Be), incorporating explicit future tidal boundaries enabled the algorithms to accurately capture the non-linear interactions between tidal surges, stochastic rainfall, and reservoir outflow. This integration significantly minimized error metrics (including RMSE) while maintaining high computational efficiency compared to traditional physical hydrodynamic models. This outcome successfully validates the hypothesis that forecasted tidal information is

a prerequisite for optimal predictive capabilities in such environments.

The primary scientific contribution of this work firmly establishes that providing ML models with a deterministic future tidal baseline fundamentally enhances their capacity to map complex estuarine hydrodynamics. In practice, this cost-effective and highly precise framework serves as a vital operational tool for flood management in vulnerable coastal metropolises like Ho Chi Minh City. It provides actionable lead times essential for proactive traffic coordination, the operation of tidal sluice systems, and the mitigation of economic losses caused by recurrent urban inundations.

Despite these advancements, the current study is constrained by the dataset's time frame of 2019 to 2022. This period may not fully capture low-frequency extreme flood anomalies or recent morphological alterations in the riverbed due to rapid urbanization. Future research should address these limitations by expanding the temporal scope, integrating projected climate change scenarios, and testing the framework across diverse global river basins to further evaluate its spatial generalizability. Ultimately, deploying this hybrid model within real-time warning systems will meaningfully advance sustainable flood response strategies in Vietnam and similar coastal regions worldwide.

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Author contributions

Ngoc Anh Le: Conceptualization, Methodology, Coding, Writing – original draft preparation. **Hao Quang Nguyen, Phong Nguyen Thanh:** Methodology, Data curation, Writing – original draft preparation. **Huy Anh Nguyen, Dao Anh Trung Le:** Visualization, Investigation. **Duong Tran Anh:** Supervision. **Son T. Mai, Bang Tran, Duong Tran Anh:** Methodology, Writing – reviewing and editing.

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No potential conflict of interest was reported by the author(s).

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Appendices

Appendix A

Machine learning models

Multilayer perceptron (MLP)

MLP is a type of artificial neural network (ANN) with a feedforward architecture, designed to model complex non-linear relationships in data. MLP was developed from the simple perceptron by Rosenblatt (1958) and extended to a multilayer architecture by Rumelhart *et al.* (1986). MLP consists of at least three layers: an input layer, one or more hidden layers, and an output layer, with neurons fully connected through weights. Non-linear activation functions (such as sigmoid, tanh, or ReLU) in the hidden layers allow MLP to represent complex data patterns. The training process uses the backpropagation algorithm to optimize weights via gradient descent, minimizing prediction errors. In this work, MLP is used to train the tidal wave frequency features obtained from Fourier analysis.

Random forest (RF)

RF is an ensemble model based on decision trees, utilizing bagging to construct multiple trees from random sample data, then aggregating the results (via averaging or voting), as proposed by Breiman (2001). At each splitting node, RF randomly selects a specific subset of features, which helps reduce overfitting and enhance the ability to handle noisy data. The results are aggregated by averaging (for regression) or majority vote (for classification). The tree diagram illustrating the structure with Yes/No branches is presented in Fig. A1(a). RF is particularly effective in forecasting rainfall-runoff models and flood management due to its capability to handle non-linear and highly noisy data (Li *et al.* 2020a, Le *et al.* 2025a).

Extreme gradient boosting (XGB)

XGB is a sequential boosting model, proposed by Chen and Guestrin (2016), which optimizes the loss function using gradient descent, and supports parallel processing and regularization to enhance performance and reduce overfitting. The trees are built sequentially, with each tree correcting the errors of the previous one, supporting parallelization, and suitable for large, complex data. The results are aggregated by weight based on the degree of error reduction. The tree diagram with Yes/No branches is illustrated in Fig. A1(b). Unlike RF, XGB prioritizes error optimization, providing high accuracy in applications such as rainfall-runoff model forecasting and flood management (Le *et al.* 2025a).

Catboost regression (CBR)

CBR, proposed by Dorogush *et al.* (2018), is a sequential boosting model that uses oblivious decision trees, where each splitting level uses the same feature and threshold, helping to increase prediction speed and reduce overfitting. CBR automatically handles categorical features without manual encoding, thanks to the “ordered boosting” technique. The trees are built sequentially, with results aggregated by weights based on the degree of error reduction. The tree diagram with Yes/No branches is illustrated in Fig. A1(c). Unlike RF (bagging) and XGB (regular boosting), CBR is optimized for categorical data, providing high efficiency in rainfall-runoff model forecasting and flood management (Le *et al.* 2025a).

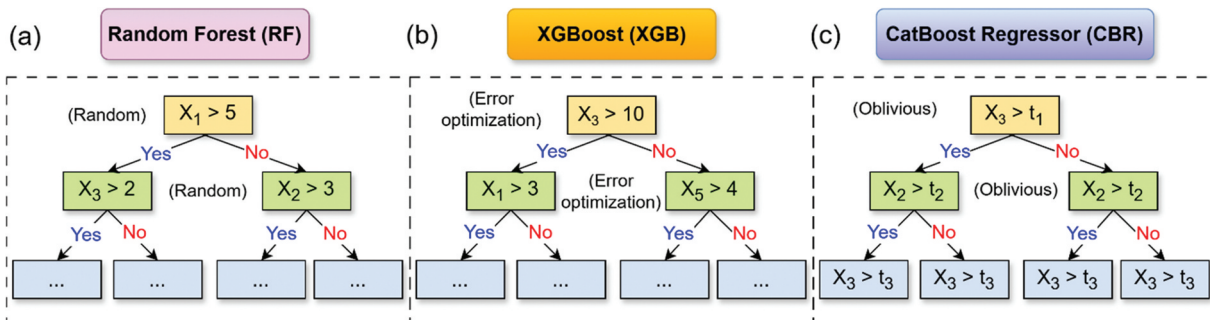


Figure A1. Decision tree diagrams showing the structures of the RF, XGB, and CBR models.

Appendix B

Table B1. Optimal hyperparameter settings for each forecasting model at the stations.

Hyperparameter	MLP	M2-rf	M2-xgb	M2-cbr	M3-rf	M3-xgb	M3-cbr
<i>Station: TDM</i>							
hidden_layer_size	(8,7)						
n_estimators		700	900	900	1900	1400	200
max_depth		4	4		8	4	
min_samples_split		10			10		
min_child_weight			10			10	
deep				3			3
learning_rate				0.01			0.01
<i>Station: PHUAN</i>							
hidden_layer_size	(64,32)						
n_estimators		1300	500	1000	800	1100	2000
max_depth		4	4		8	4	
min_samples_split		20			10		
min_child_weight			10			10	
deep				3			3
learning_rate				0.01			0.01
<i>Station: NHABE</i>							
hidden_layer_size	(8,7)						
n_estimators		500	700	1000	2000	700	2000
max_depth		4	4		8	5	
min_samples_split		20			10		
min_child_weight			10			7	
deep				3			4
learning_rate				0.01			0.01
<i>Station: VUNGTAU</i>							
hidden_layer_size	(64,32)						
n_estimators		1100	600	700	2000	1400	2000
max_depth		4	4		8	4	
min_samples_split		20			10		
min_child_weight			10			6	
deep				3			4
learning_rate				0.01			0.01